Mini Project Proposal: Workplace Compliance Advisory Desk

Team: PES1UG24AM373, PES1UG24AM374, PES1UG24AM378, PES1UG25AM817

Course: ML System Design and AgentOps (UE24AM342AA5), Mini Project 09

1. Project overview

An agentic service that answers HR compliance questions on gratuity, maternity benefit and workplace harassment under India's Labour Codes. It retrieves the provisions in force, drafts a cited advisory, verifies every citation with an independent agent, and escalates anything contested or unverifiable to legal counsel. It never gives a legal opinion.

2. Problem statement

The Labour Codes took effect on 21 November 2025 and repealed the Payment of Gratuity Act, 1972, yet most guidance still cites the old Act. Exceptions sit in provisos apart from the rules they qualify, so a single embedding search misses them: an 18-month fixed-term employee looks ineligible under the main clause, but a proviso makes gratuity payable.

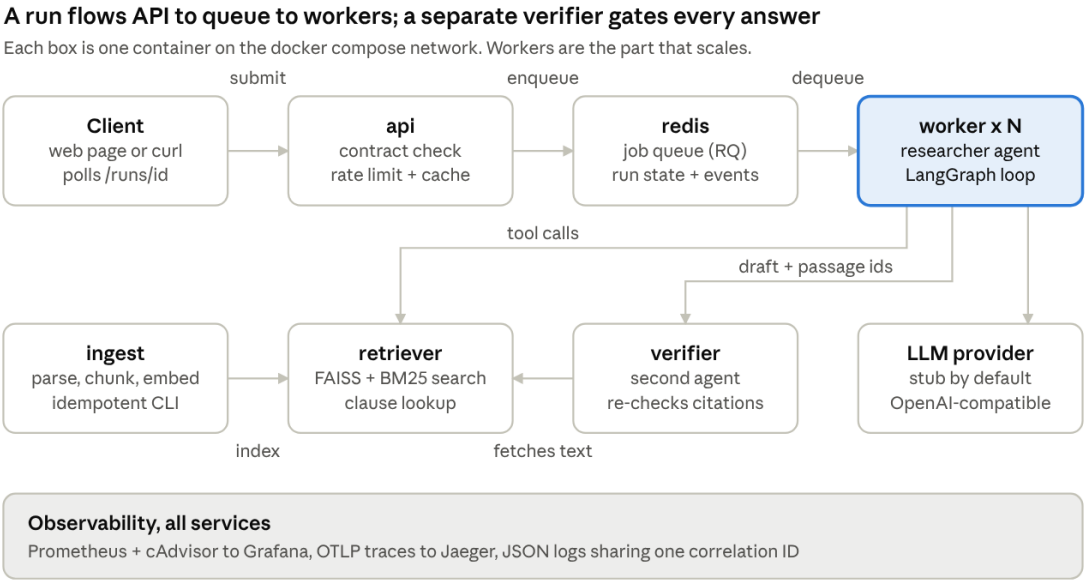
3. Proposed solution

A researcher agent searches, follows each rule to its provisos and supersession links, and drafts an answer with citations. A separate verifier service re-fetches the cited text by id and rejects any answer that cites a repealed provision, an unread proviso or unsupported text. Escalation rules are plain code, not model judgement.

4. Technology stack

Frontend	Backend	Database	AI/ML models	Frameworks
Single-page web client	FastAPI, RQ workers, Redis	SQLite chunk store, FAISS index	Local sentence-transformer embeddings; any OpenAI-compatible LLM (stub for tests)	LangGraph, LangChain, Docker Compose, Prometheus, Grafana, Jaeger

5. System architecture



architecture - 8 containers plus observability

The API validates and queues; any worker takes the job. The verifier fetches passage text itself, so a fabricated quote cannot pass.

6. Working methodology

1. **Validate:** the API checks the request against the data contract, quarantines bad input and queues a run.
2. **Guard and classify:** injection and PII guardrails run first, then the question is classified and its entities extracted.
3. **Research:** the agent calls search, get_clause and list_modifiers; a forced expand step follows supersession and rule-proviso links.
4. **Draft and verify:** a cited advisory is drafted; the verifier re-fetches each passage and checks it is in force, supported and complete.
5. **Route:** pass means answer with citation; a failed check triggers one repair, then escalation to legal counsel.

7. Key features

- **Exception-aware retrieval:** provisos are chunked on their own and linked to their rules; hybrid FAISS and BM25 search.
- **In-force awareness:** every provision carries a status and the as-of date decides what is current.
- **Independent verification:** a second service blocks unsupported or repealed citations before anything is returned.
- **Knowing when not to answer:** deterministic escalation rules for missing facts, disputes and out-of-scope questions.

- **Production operability:** async queue, scalable workers, retries and circuit breaker, tracing, metrics and cost per step.

8. Expected outcome

The brief's section 4.1 case returns gratuity payable, citing the second proviso to s.53(1), without relying on the repealed Act. On the 20-scenario set (stub model, seed corpus) task success rose from 0.45 to 0.95 and hard cases from 0/5 to 5/5. Real-model quality and the official statutes are the next measurements.